INCH-POUND
MIL-DTL-24640/24E
8 November 2011
SUPERSEDING
MIL-DTL-24640/24D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, 22 AWG, TYPE TTXW (INCLUDING VARIATION TYPE TTXOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated, 22 AWG, 7 strands, in accordance with ASTM B8.
Second	-	Mica glass tape insulation 0.0029±0.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent). Conformance testing shall be in accordance with MIL-DTL-81381 prior to cabling.
Third	-	Opaque-pigmented polyimide tape 0.001 inch (50 percent overlap minimum).
Fourth	-	Opaque-pigmented polyimide tape 0.001 inch with 0.0001 inch Fluorinated Ethylene Propylene (FEP) (50 percent overlap minimum).
Fifth	-	Modified aromatic polyimide resin coating 0.0005 inch minimum (0.001 inch nominal). Identification codes applied using Method 3 (telephone color code) or 6. (Note for Method 6, twisted pair identification code colors shall be one black and one white conductor). For Type TTXW (OW) 1½, colors shall be one white, one black, and one red conductor.
Sixth	-	Two conductors twisted together. (Except for Type TTXW (OW) 1½, shall be 3 conductors twisted together to form a triad.) The pitch shall be 1½±¼ inch.
Seventh	-	The required number of component wires shall be cabled together. Fillers may be used, as necessary, to form a round cable.
Eighth	-	Optional binder tape.

Construction Variant for Type TTXW

Ninth	-	Cross-linked polyolefin jacket (see table I for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

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Construction Variant for Type TTXOW

Ninth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.
Tenth	-	Optional binder tape.
Eleventh	-	Cross-linked polyolefin jacket (see table II for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

TABLE I. Construction details.

PIN M24640/24	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UN	TTXW-1½	1½	0.028	0.181	0.195	26.0
-02UN	TTXW-3	3	0.038	0.285	0.307	62.3
-03UN	TTXW-5	5	0.037	0.331	0.357	91.0
-04UN	TTXW-10	10	0.041	0.456	0.492	172.0
-05UN	TTXW-15	15	0.044	0.527	0.569	235.0
-06UN	TTXW-20	20	0.044	0.577	0.621	285.0
-07UN	TTXW-30	30	0.044	0.684	0.738	417.0
-08UN	TTXW-40	40	0.047	0.790	0.852	544.0

TABLE II. Construction details - overall shield.

PIN M24640/24	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UO	TTXOW-1½	1½	0.031	0.253	0.273	67.9
-02UO	TTXOW-3	3	0.038	0.333	0.359	108.0
-03UO	TTXOW-5	5	0.038	0.376	0.406	134.0
-04UO	TTXOW-15	15	0.044	0.556	0.600	299.0
-05UO	TTXOW-20	20	0.044	0.614	0.662	371.0
-06UO	TTXOW-30	30	0.047	0.717	0.772	497.0
-07UO	TTXOW-40	40	0.047	0.823	0.887	631.0

REQUIREMENTS:

Qualification required.

Construction details: See tables I and II.

Finished wire: For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-DTL-81381.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level

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Blocking	Oven temperature, 150±2 °C (302±3.6 °F) no sticking of conductors
Flammability	3 seconds (max.) after flame
	3.0 inches (max.) flame travel
	No flaming of tissue paper
Humidity resistance	5 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Impulse dielectric test	100 percent test, impulse voltage at 8.0 kV
Insulation resistance	2500 megohms for 1000 feet (min.)
Polyimide cure test	No cracking
Shrinkage	0.062 inch (max.) at 180 °C
Surface resistance	5 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.031 inch
Wet dielectric test	2500 volts (root mean square)

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	17.54
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield (TTXOW only)	1000
Shield to shield	Not applicable
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	2500
Conductor to shield (TTXOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield)	Not applicable
Conductor and shield continuity (Type TTXOW only)	No failure
Jacket flaws (Type TTXOW only)	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	Not applicable
Capacitance unbalance (percent, max.)	Not applicable

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Characteristic impedance (at 1 MHz, ohms)	Not applicable
Watertightness (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	No failure
Cross-link proof test (percent, max.)	
Jacket	50
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	No failure
Group C	
Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, Method 1 only, min.)	25
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability	No failure
Shrinkage	No failure
Surface transfer impedance (Type TTXOW only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Group D:	
Flame propagation	No failure

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Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	No failure
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)	
TTXW-1-1/2, TTXW-3, TTXW-5, TTXW-10, TTXW-15, TTXW-20, TTXW-30, TTXW-40, TTXOW-1-1/2, TTXOW-3, TTXOW-5, TTXOW-10, TTXOW-15, TTXOW-20, TTXOW-30, TTXOW-40	1½ - 15	500
	20 - 40	250

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-099)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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